Beta reduction and physical clearance in a parametric booster-to-storage-ring transfer line

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Abstract

Lower intermediate beta does not necessarily improve transfer-line clearance when dispersion, trajectory response or an immutable boundary controls the beam. For a parametric 21.789 m, 4 GeV booster-to-storage-ring line, we derive the fixed entrance ceiling on minimum five-sigma clearance and compare matched reference, low-beta and direct-clearance settings. Interior sampling corrects missed magnet extrema; refined optimization reduces vertical peak beta from 51.27 to 28.44 m. The original comparison has 406 of 600 paired shots limited by the entrance in both settings. A fresh 200-machine, three-shot ensemble finds no resolved baseline low-beta clearance benefit. At explicitly hypothetical horizontal and vertical restrictions, its paired changes are -0.152 and +0.073 mm, while direct nominal clearance optimization gives +0.881 and +0.212 mm with pointwise 95% intervals excluding zero. Station decomposition explains the competition between beam size and trajectory response. Centred transport and nonlinear Gaussian quadrature check clearance accuracy below one micrometre in representative cases. Particle tracking is reported separately: larger mean clearance does not ensure smaller observed loss counts. These results provide a conditional design criterion based on the active restriction, without certifying hardware tolerances, continuous-chamber transmission or storage-ring capture.

Keywords: transfer-line optics; physical clearance; dispersion; quadrupole matching; uncertainty

1 Introduction

Diffraction-limited storage rings require control of the distribution delivered to their injection region [1]. Korea-4GSR provides a relevant 4 GeV design context [2, 3], but a numerically matched exit ellipse is only one part of transport feasibility. An intermediate beta maximum is neither a physical aperture nor a trajectory-error tolerance. Dispersion, centroid displacement, coupling and the location of restrictions also matter.

Published transfer-line studies illustrate this distinction. SLS 2.0 separates quadrupole-scan beam characterization from position/angle control and identifies septum restrictions [4]. ESRF–EBS rematching includes off-axis injection-quadrupole fields and mechanical constraints [5]. Sirius studies distinguish injector stages and incoming beam-size effects [6]. MAX IV pulsed-sextupole injection addresses downstream ring capture [7], a different boundary from the one considered here. These primary studies do not establish the particular target or chamber dimensions of our model.

We test when lowering vertical peak beta increases minimum projected clearance, and compare that strategy with direct nominal clearance optimization at a fixed exit match. A fixed entrance can cap the minimum regardless of downstream optics; an internal restriction instead exposes competition between betatron size, dispersion and trajectory sensitivity. We make that mechanism

explicit and test it in independently seeded machine/shot ensembles. All restricted cases are parametric demonstrations, not measured chamber specifications. The BTS sequence, archived match and historical simulations are shared with the companion NKM study; this article ends at the BTS exit and claims no ring capture or kicker transparency. Standard optimization is used without algorithmic novelty claims.

2 Parametric model and physical observables

2.1 Boundaries, coordinates and source limitations

The Accelerator Toolbox (AT)/pyAT model [8, 9] contains 36 elements over 21.789 m, from marker m1 to marker m4 downstream of sept_ex. It has nine 0.2 m quadrupoles in three triplets, three main bends, an extraction kicker and two model septa (Fig. 1). Local coordinates are $(x, p_x, y, p_y, \delta, ct)$ with positions/path length in m, normalized canonical momenta approximately angles in rad, and $\delta = \Delta p/p_0$ dimensionless. The six design bend angles are $+7.5, +88.5, -111.701, +176, -111.701, +88.5$ mrad; reference geometry is held fixed when fields or particle momenta change.

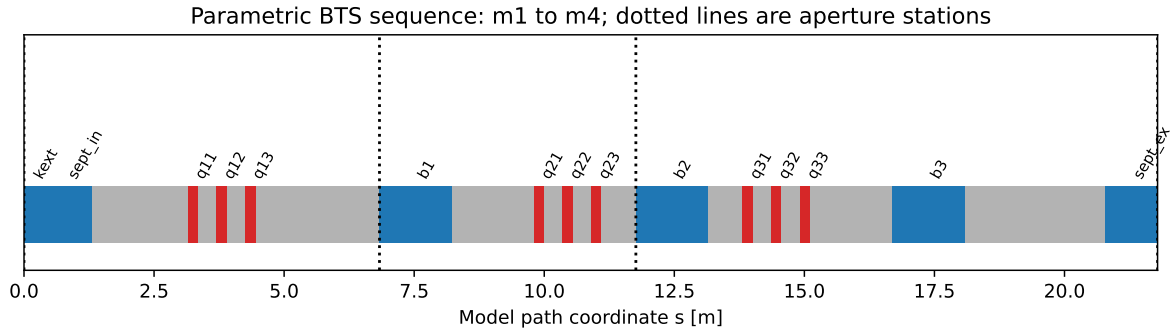


Figure 1: Configured BTS element sequence. Red blocks are quadrupoles, blue blocks are bending elements and grey blocks are drifts. Dotted lines mark the four model aperture stations. The path coordinate follows the reference trajectory; this is not a surveyed mechanical layout.

Table 1: Configured entrance and exit targets. Their physical source at these boundaries is unresolved; the exit is not the NKM-centre ring optics.

Observable	m1 entrance	m4 target
β_x, β_y [m]	7.560000, 12.269000	2.336495, 4.256241
α_x, α_y [1]	1.523100, −1.654700	−0.016335, 0.017772
D_x [m], D'_x [1]	0.276200, −0.065700	0.080868, 0.047472

Entrance and target functions (Table 1) are inherited configured values, treated as parametric. No transport from m4 to a ring matching point is established. Quadrupole family bounds, bore radius, pole-field limit, the 60 m beta screening ceiling and square aperture limits are modelling assumptions, not verified hardware specifications. The legacy MAD-X line has ten active quadrupoles, different drift/bend/edge definitions and an unused additional quadrupole definition. It is not an equivalent independent lattice; our independent validation implements the identical AT sequence analytically.

Aperture elements have half-widths 19.35 mm in both planes at $s = 0$ and 21.789 m, and 30 mm at 6.833 and 11.761 m. The continuous chamber and septum channel restrictions between

them are unknown. Station transmission therefore cannot certify machine transmission; no conservative bound for the unknown chamber is claimed.

2.2 Beam covariance and clearance

The new study uses the booster horizontal design input $\epsilon_x = 7.67$ nm rad and $\sigma_\delta = 0.106\%$ [3]. The vertical input $\epsilon_y = 0.767$ nm rad is an assumed 10% ratio. Betatron coordinates and momentum are independent untruncated Gaussians before dispersion is added; initial transverse coupling is zero and $ct = 0$ for this RF-free single pass. The transverse blocks of the complete covariance are

$$\Sigma_{u,\mathrm{bet}} = \epsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & (1+\alpha_u^2)/\beta_u \end{pmatrix}, \qquad \Sigma_0 = \Sigma_{\mathrm{bet}} + \sigma_\delta^2 \mathbf{d}\mathbf{d}^\mathsf{T}, \tag{1}$$

where $\mathbf{d} = (D_x, D_x', D_y, D_y', 1, 0)^\mathsf{T}$. Rolls can generate coupling downstream. Full covariance transport, $\Sigma_s = M_s \Sigma_0 M_s^\mathsf{T}$, retains that coupling and momentum correlation. A local affine map gives the centroid $\bar{\mathbf{z}}_s = M_s \bar{\mathbf{z}}_0 + \mathbf{b}_s$.

For each aperture plane with lower/upper edges a_-, a_+, we define

$$c_u(s) = \min[a_+ - \bar{u}(s), \bar{u}(s) - a_-] - 5\sqrt{\Sigma_{uu}(s)}, \qquad c_{\min} = \min_{s\in\text{stations},u\in\{x,y\}} c_u(s). \tag{2}$$

Five sigma is a declared projected coverage metric, not complete containment of an unbounded Gaussian. Beta reduction in m must not be called a clearance increase in m. Historical optics-only ensembles use a separate conservative 100/10 nm rad, 0.11% beam configuration and are not pooled with this study.

At 4 GeV the rigidity is approximately 13.34256 T m. For K in m^{-2}, $G = K(B\rho)$ in T/m and $B_{\mathrm{pole}} = |G|r$ in T. Assumed bounds are $|K| \le 2.5$ for q11/q33 and $|K| \le 3$ for the other quadrupoles. With the assumed 19.35 mm bore, the respective maxima are 33.356/40.028 T/m and 0.645/0.775 T, below the assumed 1.2 T screening limit. This calculation checks model consistency, not magnet engineering feasibility.

3 Physical criterion, validation and comparison protocol

3.1 Active restrictions and the entrance ceiling

The full minimum satisfies

$$c_{\min}(K) = \min[c_{\mathrm{entrance}}, c_{\mathrm{downstream}}(K)].$$

Because the entrance precedes adjustable quadrupoles, its clearance is independent of K. Therefore $c_{\min} \le c_{\mathrm{entrance}}$, with equality whenever every downstream clearance exceeds that ceiling. The metric then saturates: lowering an intermediate beta cannot increase its nominal value. This is a sufficient condition for saturation, not a claim that downstream profiles are unchanged.

At a single symmetric active station away from $\bar{u} = 0$, the local sensitivity is

$$\frac{\partial c_u}{\partial K_j} = -\operatorname{sgn}(\bar{u})\frac{\partial \bar{u}}{\partial K_j} - \frac{5}{2\sigma_u}\frac{\partial \Sigma_{uu}}{\partial K_j}. \tag{3}$$

Switching restrictions or crossing zero centroid makes this expression piecewise. In an uncoupled plane $\Sigma_{uu} = \epsilon_u \beta_u + D_u^2 \sigma_\delta^2$; our coupled calculation retains the complete covariance. Reducing beta controls one size contribution, not necessarily dispersion or orbit sensitivity. We export these variances and centroids separately, adding variances rather than RMS sizes.

3.2 Accuracy and spatial observation resolution

Exact independent hard-edge maps are checked against AT tracking at every element boundary. The original ten-step integrator misses its 2×10^{-6} nominal maximum-entry tolerance. Twenty/forty/eighty steps give discrepancies 2.53×10^{-7}, 1.58×10^{-8} and 9.89×10^{-10}; new studies use forty. Transverse symplectic discrepancy is 1.22×10^{-15}, without a longitudinal six-dimensional certificate. Translations, rolls and fixed-geometry dipole-field perturbations are exercised independently. Momentum errors enter particle δ without double rescaling magnet fields.

The original affine error maps omit chromatic derivatives about displaced trajectories. Comparing all 600 original shots with AT Jacobians expanded about the actual centroid changes station clearances by up to 78.43 micrometres, but shifts the paired mean by only -0.155 micrometres. Eight reference and three alternative active-station classifications change. The original mean 14.61 micrometres and interval $[-17.91, +48.32]$ become 14.45 and $[-17.88, +48.10]$; the unresolved-benefit conclusion persists. New comparisons use centred AT maps. Moment probes are uncensored by aperture limits, so propagated moments describe the incoming distribution rather than a survivor-biased subset.

Representative simultaneous-error cases compare probe steps 10^{-6}–10^{-8} and forty/eighty integration steps. Their clearance changes are below 5×10^{-6} micrometres. Product Gauss–Hermite quadrature at orders three, five and seven propagates the unrestricted Gaussian through nonlinear AT; five/seven convergence is negligible on the checked cases. Centred-linear versus nonlinear-moment discrepancy is at most 0.606 micrometres on four original extreme or representative shots and 0.6858 micrometres on two new machines for every regime/design. These finite-case checks are smaller than the reported effect intervals and illustrative 0.1 mm decision scale; they are not universal error bounds or Gaussian loss-tail certificates. Intentional scraping verifies first-loss accounting. Detailed checks are supplied in the supplement/evidence.

Spatial observation is refined independently of integration steps. Exact maps sample each finite element at 8, 16, 32 and 64 subdivisions, applying bend edges only once. Original reference boundary peaks 31.44/50.75 m become 31.63/51.27 m; old low-beta peaks 33.58/28.18 m become 33.77/28.50 m. Reoptimizing vertical peak beta on sixteen subdivisions converges in 34 SLSQP iterations. The resulting 64-subdivision peaks are 33.79/28.44 m; changes from 32 subdivisions are below 0.00035 m. These are spatially converged sampled maxima, not exact analytical extrema or a global optimum. q32 remains at its assumed lower bound $-3\ \mathrm{m}^{-2}$.

3.3 Frozen strategies and hypothetical restrictions

All settings share entrance/target optics, quadrupole bounds and beam assumptions. Six exit residuals are scaled by $(0.05\,\mathrm{m}, 0.05\,\mathrm{m}, 0.01, 0.01, 0.002\,\mathrm{m}, 0.001)$. Nominal optimization imposes six residual equalities and horizontal peak beta ≤ 60 m. Direct clearance uses a millimetre-scaled epigraph: maximize t subject to $t \leq c_u(s)$ at every station/plane. Two starts (reference and refined low beta) use 200-iteration budgets and 10^{-8} termination tolerance. All six searches converge; their iterations, settings, active bounds and independently checked feasibility are supplied. Exit residuals satisfy declared scales and horizontal screening is checked at 64 subdivisions. Entrance-ceiling ties choose the smallest reference-setting change; their tiny numerical effects are not interpreted as physical gains.

Three predeclared regimes keep entrance/exit limits unchanged: baseline; a horizontal 3 mm halfwidth only at m2 ($s = 6.833$ m); and a vertical 1.5 mm halfwidth only at m3 ($s = 11.761$ m). Other limits remain unchanged. These narrow internal restrictions are deliberate hypothetical stress cases, not surveyed hardware. A nominal halfwidth sweep of 1.5, 2, 3, 5 and 30 mm displays changes in the active restriction without using test outcomes for design selection. The legacy lattice shares an aperture object at its internal stations; the new driver detaches the selected station before editing its limit. An invalid initial run that inadvertently narrowed both stations is

retained separately and excluded.

Settings are frozen before a new independently seeded test. Static errors are independent Gaussian quadrupole relative gradients 0.1%, x/y offsets 100 micrometres, rolls 0.5 mrad and dipole relative fields 0.05%. Each of 200 machines has three independent shots with 5% entrance beta fluctuations (alpha fixed), horizontal position RMS 0.5 mm, angle RMS 0.2 mrad and centroid momentum RMS 0.1%. These error levels remain modelling assumptions. Static, shot and particle seeds are 82002, 82003 and 82004, respectively. Each shot has 2,000 untruncated Gaussian particles with the complete design-beam covariance; all nine regime/strategy combinations receive common errors and particle draws. No optical correction or per-shot retuning is used. The total is 10.8 million particle transports, 1.2 million per combination, not 10.8 million machines.

The primary contrast is alternative minus reference in machine-average minimum five-sigma clearance. Ten thousand paired machine-cluster bootstrap replicates (seed 82005) give pointwise descriptive 95% percentile intervals. No simultaneous coverage claim is made across strategies/regimes. The predeclared useful-effect scale 0.1 mm is illustrative, not a hardware requirement. First loss is nonfinite x after an element; all six coordinates of survivors must be finite. Element exit observations and first-loss indices are retained. Transmission is a separate observable. No additional samples were added after inspecting outcomes.

4 Results

4.1 Saturation and heterogeneous original outcomes

At zero error, both original settings have minimum clearance 17.455 mm at the entrance (Table 2). The full station audit reconstructs all 600 saved paired minima within 10^{-12} m. Every one of the 406 exact shot ties has the horizontal entrance active in both designs. The reference has 449 entrance-x, 103 exit-x and 48 exit-y limiting shots; the old low-beta setting has 467, 113 and 20 (Fig. 2). Neither internal station limits that original ensemble.

Table 2: Original nominal station clearances [mm], before spatial reoptimization. R denotes reference and L the old low-beta setting. The entrance caps the full minimum.

s [m]	$c_{x,R}$	$c_{x,L}$	$c_{y,R}$	$c_{y,L}$
0.000	17.455	17.455	18.865	18.865
6.833	27.518	27.416	29.258	29.388
11.761	28.123	28.129	29.117	29.320
21.789	18.555	18.555	19.064	19.064

The 200 original machine averages include 54 gains, 53 losses and 93 exact ties. Difference standard deviation is 0.236 mm, with 5th/median/95th percentiles $-0.322/0/+0.403$ mm and range -0.923 to $+1.401$ mm. Its near-zero mean therefore conceals offsetting responses. Excluding the entrance gives a secondary downstream-only mean difference $+0.056$ mm; this diagnostic does not replace the original full-minimum estimand. Historical optical-rule failures, matching replay and local response diagnostics are retained in the supplement.

4.2 Aperture-dependent strategy outcomes on new machines

Figure 3 shows the spatially refined beta profiles. The fresh baseline low-beta change is 0.0103 mm with interval [-0.0156, 0.0377] mm, again unresolved. This does not prove exact equivalence or universal inferiority of low-beta design.

At the horizontal restriction, low beta worsens the mean by 0.152 mm, while direct nominal clearance gains 0.881 mm (Table 3). Nominal horizontal betatron RMS changes from 0.220 to 0.230 mm in the direct design, but dispersive RMS decreases from 0.445 to 0.292 mm. On the

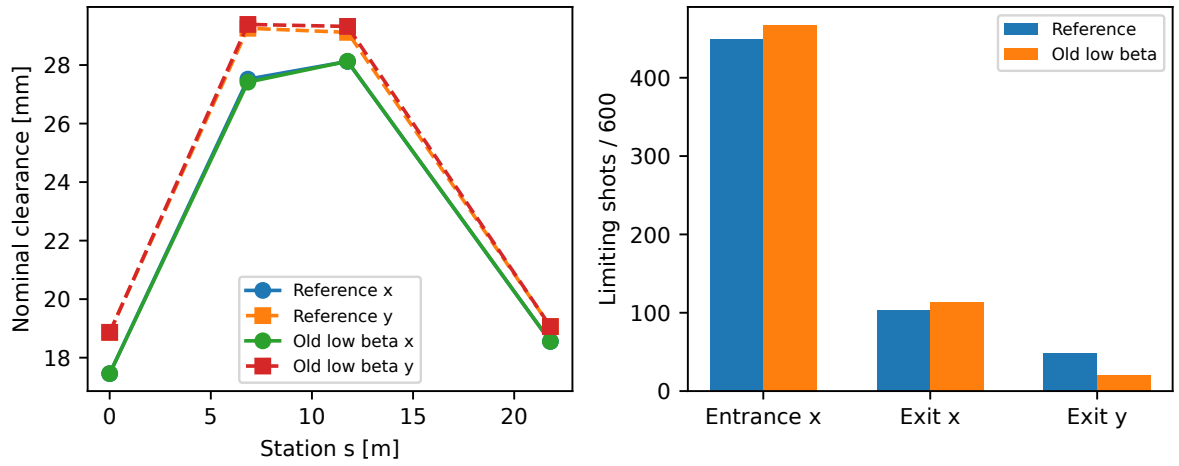


Figure 2: Original affine-model audit. Left: nominal station clearances for both settings and planes. Right: active-limit counts in 600 paired shots, exposing entrance saturation and exit switches. These retrospective data are separate from the new test and refined setting.

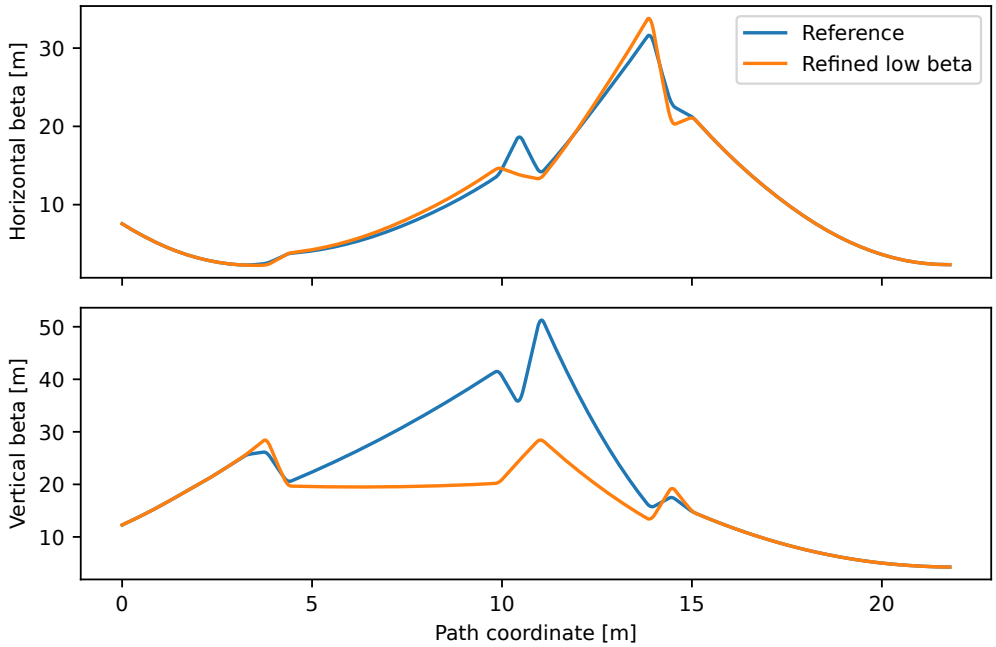


Figure 3: Reference and refined low-beta profiles evaluated inside magnets with 64 spatial subdivisions per finite element. The exit match remains fixed; the vertical improvement also changes horizontal transport.

Table 3: Frozen-strategy comparisons. Nominal minimum and new paired mean change are in mm; brackets give pointwise 95% machine-cluster intervals. Restricted cases are hypothetical. The baseline clearance-setting change is numerical.

Regime	Strategy	Nominal	Mean change	95% interval
Baseline	low beta	17.455	+0.010	[-0.016, +0.038]
Baseline	clearance	17.455	–	degenerate
x restriction	low beta	0.420	-0.152	[-0.157, -0.147]
x restriction	clearance	1.142	+0.881	[+0.847, +0.914]
y restriction	low beta	0.821	+0.073	[+0.048, +0.098]
y restriction	clearance	0.915	+0.212	[+0.174, +0.250]

held-out shots, mean absolute horizontal centroid decreases from 1.218 to 0.960 mm and mean five-RMS size decreases from 2.485 to 1.862 mm. Their combined changes explain the gain; minimizing vertical beta alone does not control either term.

At the vertical restriction, low beta gains 0.073 mm and direct clearance gains 0.212 mm. Low beta reduces mean five-RMS size from 0.885 to 0.681 mm, but increases mean absolute centroid from 0.230 to 0.360 mm; much of its nominal size benefit is offset by trajectory sensitivity. Direct clearance gives 0.586 and 0.316 mm for these terms. The low-beta interval excludes zero but lies below the illustrative 0.1 mm useful-effect scale; direct gains exceed that scale in both restricted cases. These are outcomes for the selected finite searches and assumptions, not universal strategy dominance.

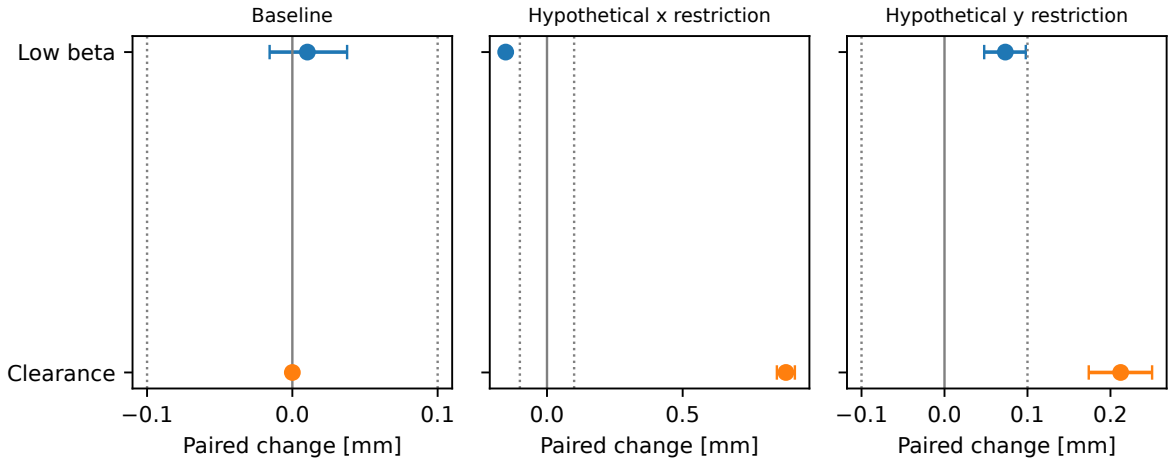


Figure 4: New paired mean clearance changes with pointwise 95% intervals for the 200 independent machines. Dotted vertical lines indicate the declared illustrative ±0.1 mm scale. Every regime includes the entrance and exit in its minimum.

4.3 Physical particle loss is a separate outcome

No baseline losses occur. For zero of 200 independent blocks, the one-sided 95% upper probability of an observed-loss block is 1.4867% [10]. The event consists of three sampled shots with 2,000 particles per shot; it is not the probability of any loss during indefinite operation. Shared machine and shot randomness prevents interpreting all particles as independent machine reliability trials. The supplement gives exact block intervals and separately labelled descriptive particle bounds.

The vertical restriction illustrates that a higher mean clearance need not produce fewer losses: reference/low-beta/direct settings lose 1/43/26 particles in 1/6/2 blocks (Table 4). These few tail

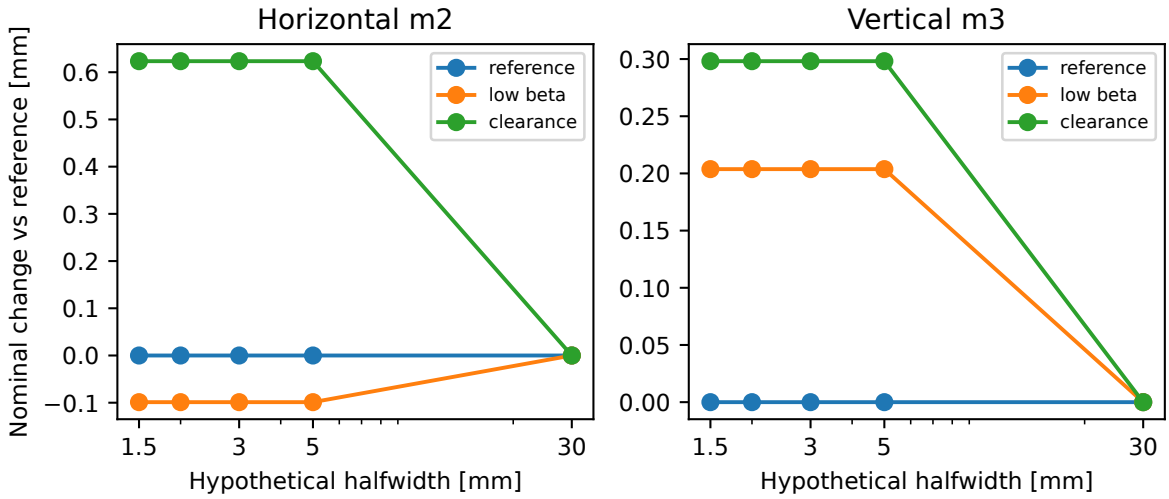


Figure 5: Nominal clearance changes versus reference across halfwidth sweeps, with frozen settings. The adjusted plane is horizontal m2 or vertical m3; all other limits stay fixed. A wide restriction returns to the entrance ceiling. Hypothetical sweeps expose metric saturation; they are not a prediction of chamber dimensions.

Table 4: Observed first losses in the new test: 1.2 million particles and 200 three-shot machine blocks per row. A block is affected if any sampled particle is lost. Counts refer to model stations, not continuous-chamber or ring losses.

Regime	Strategy	Lost particles	Affected blocks
Baseline	reference	0	0
Baseline	low beta	0	0
Baseline	clearance	0	0
x restriction	reference	65798	155
x restriction	low beta	79736	163
x restriction	clearance	17200	72
y restriction	reference	1	1
y restriction	low beta	43	6
y restriction	clearance	26	2

events do not establish a converged machine reliability ranking. In the stronger horizontal stress case, loss counts and affected blocks are lower for direct clearance than reference, and higher for low beta. Paired transmission intervals are supplied separately; clearance intervals must not be reused as loss-probability statements.

5 Discussion and conclusions

The physical criterion is the active restriction, not beta alone. A fixed entrance gives an immutable ceiling; an interior restriction exposes dispersion and trajectory response competing with betatron size. The paired tests provide three conditional regimes: unresolved baseline change, an adverse low-beta horizontal outcome, and a beneficial but modest vertical outcome. Direct nominal clearance settings improve the two restricted mean-clearance cases, while the particle study demonstrates that mean improvement is not tail reliability.

This adds an observable-specific comparison to established transfer-line rematching and trajectory-control practice, rather than a new optimizer. Optimization should state the limiting station, retain full covariance and centroid response, verify interior extrema, and independently evaluate its chosen observable. Robust training-based optimization, operational correction and global search are not demonstrated. Source-confirmed chamber geometry, matching boundaries and error correlations could change the preferred setting.

All targets, continuous restrictions and hardware limits remain as described in Section 2. Independent-Gaussian errors and hypothetical restrictions are parametric assumptions. Numerical precision cannot resolve their physical uncertainty. This paper neither certifies real BTS transmission nor derives construction tolerances, downstream capture or NKM transparency. The original statistical campaign is archived separately; its optical-rule failures are not particle losses. The resulting design lesson is conditional and testable with confirmed engineering inputs or future measurements.

Statements and Declarations

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Generative assistance. OpenAI Codex assisted with software, numerical execution, literature checking and manuscript drafting beyond copy editing. The author must validate and approve the final work.

Data and code availability. A local referee attachment supplies selected raw data, configurations, source snapshots, environment identities, hashes and reproduction commands. It has not been publicly deposited or sent to a journal; reviewer delivery and final availability wording remain author actions.

A Evidence and supplementary material

The supplement preserves historical matching/optical statistics and reports block-loss inference and numerical limitations. Reviewed and original campaign identities, all previous reports, new frozen designs, station decompositions and point-by-point responses are included in the local attachment. Each numerical stage records source hashes and uses a fresh result directory. The first invalid shared-aperture run is retained as a failure record, not selected evidence. Technical validation does not imply journal acceptance.

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